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NOTE

Spatial variation in porosity and skeletal element characteristics in apical tips of the branching coral *Acropora pulchra* (Brook 1891)

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Abstract Micro-CT scanning techniques were used to investigate fine-scale variation in porosity along branch tips of *Acropora pulchra*. Porosity variation is a result of progressive thickening of skeletal elements away from the apical tip of branches, rather than changes in the spacing of skeletal elements. A linear fit was found to describe the relationship between distance along the tip and both porosity and skeletal thickness. The slope of the line obtained may relate to branch extension rates and allow retrospective data to be obtained from *Acropora* specimens. Skeletal morphology examined by 2D and 3D imaging shows a progressive gradation in thickness occurring in the axial corallite wall and thickness changes at a site of incipient branch formation. The application of the micro-CT technique to museum and fossil specimens is illustrated.

Keywords Acropora · Porosity · CT scan · Skeletal structure

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Introduction

Many massive coral species display a defined regular pattern of porosity variation that corresponds to cycles in the deposition of aragonite by the layer of tissue at the outer colony surface. This variation is frequently linked to seasonal environmental changes and has been demonstrated to correlate with instrumental measurements of sea surface temperature (SST), light and rainfall (Dodge and Vaisnys 1975; Wellington and Glynn 1983; Carricart-Ganivet et al. 2007). In contrast, the broad pattern of porosity variation in branching coral species of the genus *Acropora* is a decrease from the outermost branch tips towards the base of colonies. However, the detailed spatial pattern over which this occurs is not well understood in comparison with massive corals. Skeletal structure in the *Acropora* genus is characterised by the differentiation of corallites into axial corallites primarily responsible for skeletal extension, and radial corallites that remain fully connected to the axial corallite in the branch interior (Figs. 1, 3). Associated with this skeletal arrangement, living tissue in *Acropora* species extends deep within the skeleton, and continued calcification prevents formation of a regular density banding pattern as occurs in massive species, although a form of annual radial growth banding has been documented in *A. palmata* (Gladfelter 2007).

At the colony scale, changes in *Acropora* skeletal porosity associated with distance from the apical tip have been found to vary with skeletal age in *A. cervicornis* (Gladfelter 1982) and to be either linearly related or uniform in *A. formosa* (Oliver 1984). The model for skeletal growth in *A. cervicornis* consists of differing phases of calcification, with rapid initial growth of soft tissue on a thin scaffolding of skeletal elements followed by secondary thickening of the skeletal elements (Gladfelter 1982). According to this model, the secondary infilling of the skeleton which obscures any potential environmental record does not substantially occur in the first ~5 cm of the tip in *A. cervicornis*. This feature of *Acropora* skeletal development has led to the suggestion that the initial porosity of young branch tips may be influenced by environmental conditions and that, consequently, branching *Acropora* specimens may have potential as palaeo-environmental recorders via a similar mechanism to that widely reported in massive coral species (Bucher et al. 1998).

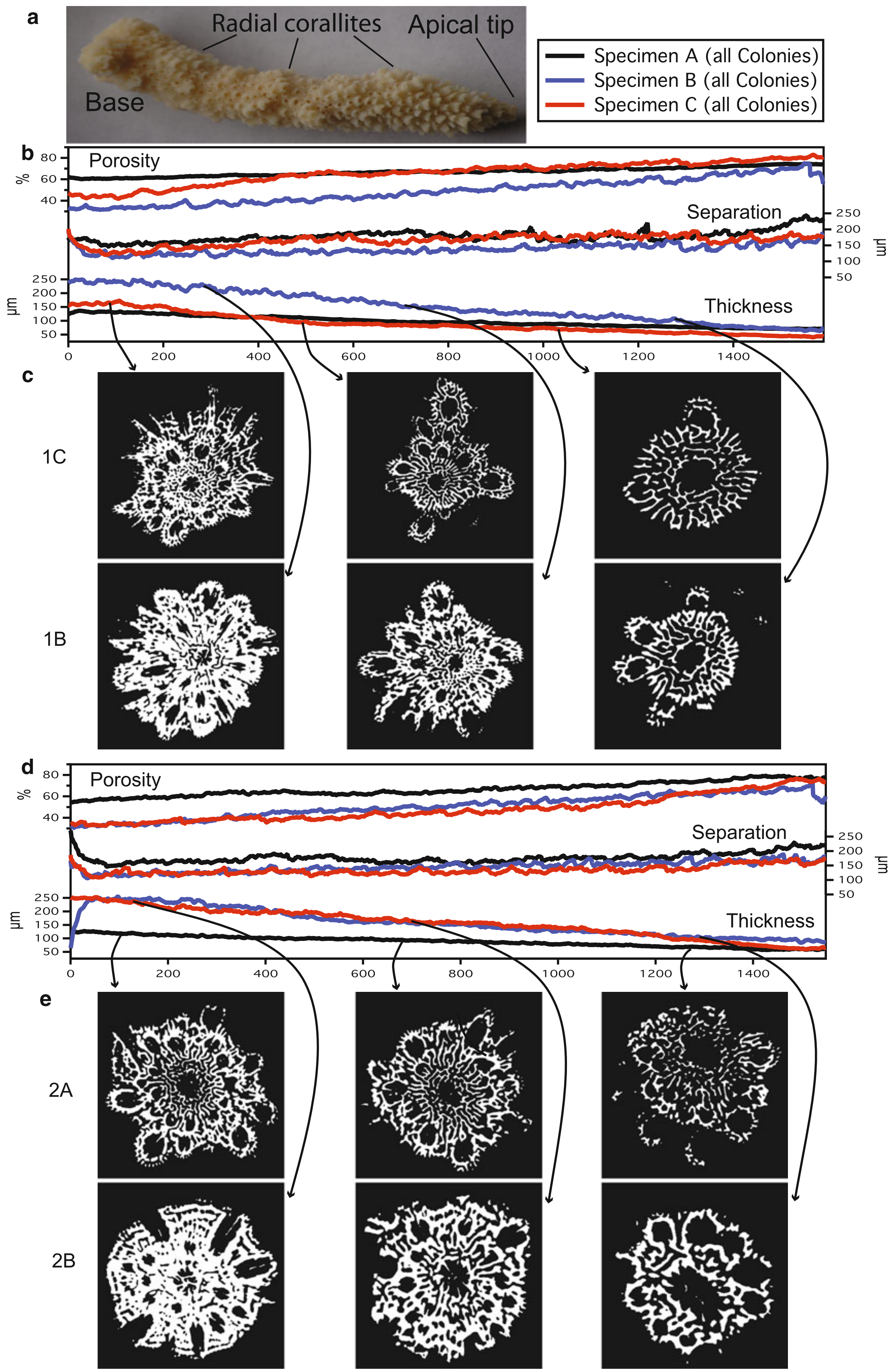
A variety of techniques are used to obtain information on spatial variations in coral skeletal porosity and structure, including X-radiography (Barnes and Devereux 1988), gamma densitometry (Chalker and Barnes 1990), Archimedean principles (Bucher et al. 1998), mercury intrusion porosometry (Laine et al. 2008) and ultrasound (Cleveland et al. 2004). These techniques provide information on the density properties of the skeleton, but do not provide direct information on the variations in skeletal structure that cause such changes. Interpretation of the secondary images produced by these techniques is often complex, and inference of the underlying changes in skeletal structure is difficult (Le Tissier et al. 1994). Direct imaging of coral skeleton has traditionally been carried out by the preparation of portions of the skeleton as petrographic thin sections or for scanning electron microscopy (SEM) analysis (Gladfelter 1982). However, these methods have the disadvantage of requiring partial or complete sacrifice of coral samples as a result of coating procedures or cutting to expose internal skeleton, and thus are unsuitable for valuable or historical specimens from museum collections. Recently, computed tomography (CT) scanning has been added to the suite of techniques available for studying coral skeleton and used for a range of applications, such as visualisation of borer activity within coral skeleton (Schönberg 2001; Beuck et al. 2007), and obtaining 3D growth rate measurements (Saenger et al. 2008). In this study, we used micro-CT scanning techniques to examine variation in porosity within branch tips of *A. pulchra* in newly collected samples, specimens from historical museum collections and Holocene material. We investigated the relationship of skeletal thickness and spacing to overall porosity and used 2D and 3D visualisation techniques to investigate changes in skeletal structure underlying porosity variation along branches from base to tip.

Fig. 1 **a** Image of *Acropora pulchra* branch specimen used in this ▶ study. All porosity, skeletal element separation and thickness plots are orientated in the same direction with respect to the base and tip as this specimen. **b** Porosity, skeletal element separation and thickness plots for colony 1. **c** An example of the skeletal morphology changes occurring within a specimen with high porosity (1C) and low porosity (1B). Images shown are at the same distance from the apical tip of each specimen. **d** Porosity, skeletal element separation and thickness plots for colony 2. **e** Further illustration of differences in skeletal morphology underlying a specimen with high porosity (2A) and low porosity (2B)

Methods

Modern *Acropora pulchra* samples used in this study were collected from King Reef (17° 46′ S, 146° 07′ E), a large shore-attached reef in the nearshore region of the central Great Barrier Reef (GBR), Australia. Three branch tips (4–5 cm length, 0.80–1.76 cm width) were removed from each of five colonies growing along a cross-reef transect in July 2008 ($n = 15$). Specimens were labelled according to colony (1–5) and branch tip (A–C). Following inspection, one specimen (5C) was excluded from analysis due to an abnormal growth morphology. A fossil specimen was obtained from reef cores collected from King Reef. Based on radiocarbon dating of corals at similar depth, the age of the fossil specimen is estimated at ~4,000 cal yBP. The museum specimen examined in this study was collected at Rocky Island Reef (12° 53′ S 143° 32′ E), a nearshore reef in the northern GBR, in 1892 by William Saville-Kent, and is stored in the Natural History Museum, London (NHMUK ZOOL 1892.6.8.38).

Micro-CT data were collected at the Natural History Museum (London, UK) with an X-Tek HMX ST225 cone beam system (Nikon Metrology, Tring, UK). The protocol used X-rays generated from a molybdenum target using a voltage and current of 80 kV and 100–120 μA, respectively. A total of 3,142 angular projections were collected at 0.11° angular intervals in a single 360° rotation. The radial projections were reconstructed into a three-dimensional matrix of isotropic voxels ranging from 20 to 30 microns, depending on the height of the coral tip. In theory, the size-related variation in voxel size (and therefore resolution) might have created a systemic measurement error; however, in practice, none was observable in the data. The authors recommend that wherever possible, users adopt a protocol that allows all specimens to be scanned at the same voxel size. It is possible to normalise voxel size after scanning but the step requires interpolating the data, which may also introduce a systematic error. The volumes were reconstructed using CT-Pro Version 2.0 (Nikon Metrology, Tring, UK) and rendered using VG Studio Max 2.0 (Volume graphics, Heidelberg, Germany). The images were segmented using a half-maximum height threshold; each


a
Radial corallites
Apical tip
Base
Specimen A (all Colonies)
Specimen B (all Colonies)
Specimen C (all Colonies)
b
Porosity
Separation
Thickness
%
µm
c
1C
1B
d
Porosity
Separation
Thickness
e
2A
2B

transverse slice of the volume was reduced to a series of linear transects for greyscale values. The greyscale value associated with the mid-point between adjacent peaks and troughs within each transect was calculated as the 'Half-Height Value' (McColl et al. 2006). Average half-height values for all transects in all slices were then used as a global threshold value for the volume.

Porosity was calculated for sequential transverse slices from the tip to base of each specimen, as the percent proportion of void within the total area of the region of interest. This was accomplished using the CTAn software package (Skyscan, Kontich, Belgium). A shrink-wrapping algorithm that mapped a spline onto each image slice was used to define the outer perimeter of the coral. Skeletal element thickness was also calculated for each transverse slice from tip to base, as the mean thickness summed over all local pixels based on the diameter of a sphere fully contained within the skeletal structure. Average values for the separation (determined by the maximum diameter of a sphere fully contained within the pore structure) of skeletal elements within each transverse slice were also obtained. Least squares regression analysis was performed between porosity and image slice number, and skeletal element thickness and slice number. To investigate changes in skeletal thickness between the distal ends and bases of sampled tips, and to illustrate changes at points of incipient branching, a 'hotspot' colour gradient was applied to specimen 2C. To visualise internal skeletal structure, and skeletal thickness and spacing variation in 3D, several animations were generated from the reconstructed volume of specimen 2C (Electronic Supplemental Material, ESM 2, 3, 4).

Results and discussion

Porosity within the *Acropora* tips decreased from tip to base in all specimens. There was no abrupt change in porosity value at any distinct point with distance from the tip of the specimen. A regression of porosity and distance from the tip of each specimen found that a straight line was an excellent fit between these two variables in all modern specimens (Table 1). Skeletal element thickness was found to increase from tip to base, and a linear fit was found to closely represent the relationship between skeletal element thickness and distance from the specimen tip. Separation of skeletal elements was also related to distance from the specimen tip, but the relationship was weaker than for porosity and thickness. The relationship between skeletal porosity and distance from the apical tip has previously been investigated in *A. cervicornis* (Gladfelter 1982). A change in the slope of the line between percent mineralization (the inverse of porosity) and distance from the apical tip was found at 30 cm, with minimal secondary thickening occurring in the first 5 cm of the tip (Gladfelter 1982). If minimal secondary infilling occurs in the first 5 cm, this suggests that porosity variations result from changes in the spacing of skeletal elements of equal thickness. However, results obtained in the present study show that porosity changes that occur in the distal few cm of a branch are primarily related to changes in skeletal element thickness occurring in this portion of the skeleton. Graphs of thickness and separation values for each slice show progressive changes in skeletal element thickness occur along the entire apical branch length, whilst the separation of elements remains relatively constant (Figs. 1, 2).

Table 1 Regression (R^2) coefficient values for the relationship between distance from the apical tip and skeletal porosity, skeletal element thickness and skeletal element separation

Specimen	Porosity r^2 value	Thickness r^2 value	Separation r^2 value
1a	0.970	0.949	0.436
1b	0.960	0.968	0.602
1c	0.916	0.821	0.537
2a	0.929	0.972	0.347
2b	0.958	0.895	0.808
2c	0.889	0.964	0.501
3a	0.908	0.926	0.157
3b	0.956	0.907	0.837
3c	0.976	0.930	0.616
4a	0.920	0.864	0.712
4b	0.893	0.849	0.658
4c	0.903	0.860	0.659
5a	0.960	0.941	0.830
5b	0.947	0.915	0.780
Rocky Island	0.803	0.953	−0.008*
Holocene	0.303	0.412	−0.064*

All values are significant at $P = <0.05$, with the exception of values marked with an *, which are not significant

There is interest in the potential of *Acropora* as a source of material for isotopic and trace element analysis due to the fact that acroporids are often the dominant coral found within palaeo-reef cores; however, the difficulty of infilling skeleton with a different chemical composition is an acknowledged limitation (Shirai et al. 2008). Oliver (1984) reported that in the absence of linear extension, secondary infilling continues in *A. formosa* to produce a constant porosity value along the branch length. The relationship between porosity and distance from the tip observed in this study suggests that both linear extension and secondary infilling processes were occurring in the scanned branch tips. The extension rate of *A. formosa* tips was shown to be inversely related to their density (Oliver 1984). The slope

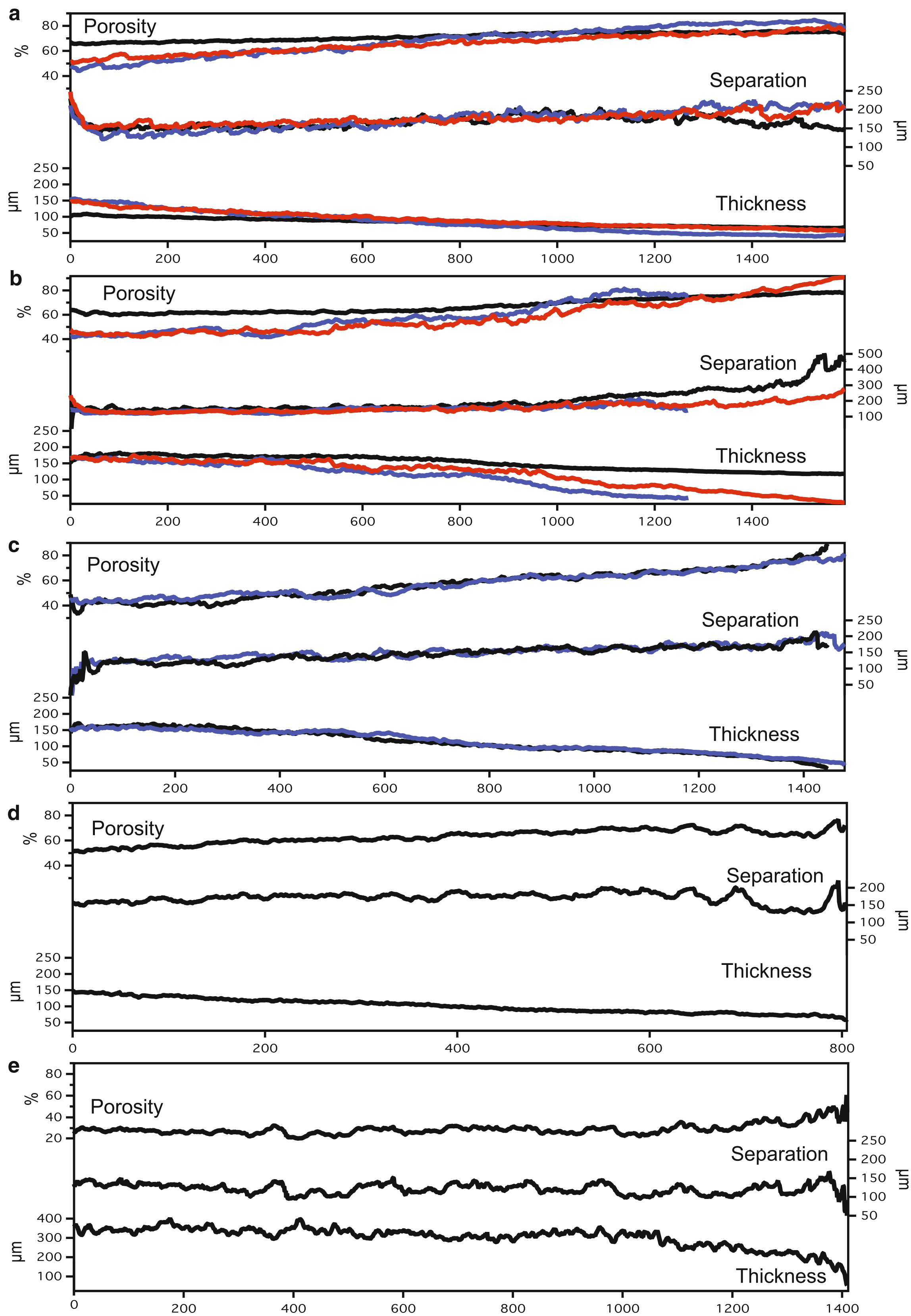


Fig. 2 Porosity, skeletal element separation and thickness of branches within remaining *Acropora pulchra* colonies. **a** Colony 3. **b** Colony 4. **c** Colony 5. **d** Rocky Island Reef specimen. **e** Holocene specimen

of the line between skeletal porosity and distance from the tip, as obtained from *Acropora* specimens, in this study is likely to correlate well with extension rate and could potentially be used for retrospective analysis of growth rates in *Acropora* species obtained from palaeo-reef cores and museum specimens, although further field and laboratory experimentation would be required to confirm this relationship. The slope of the line of thickness and porosity for the museum specimen used in this study is within the range of modern values (Fig. 1; Table ESM 1). In contrast, the fossil specimen was characterised by a thicker skeletal element structure and correspondingly lower porosity values than modern specimens. This may be indicative of diagenetic processes such as secondary aragonite precipitation, or that fossil *Acropora* growth resulted in thicker skeletal structure at this site.

3D imaging allows characteristics of skeletal structures to be visualised by application of a 'hotspot' colour gradient to show the progressive increase in skeletal element thickness from apical tip towards the base of the branch and shows skeletal thickening occurring around the site of incipient branch development (Fig. 3a, b). Little is known about the branching mechanism in *Acropora* species, although overall *Acropora* colony morphology has been shown to vary taxonomically (Wallace and Willis 1994), in relation to wave energy (Bottjer 1980), and light levels (Kaniewska et al. 2009). It has also recently been shown that growth of new axial corallites in *A. pulchra* is related to light availability and wavelength (Kaniewska et al. 2009). At the level of biomineralisation, the process by which branching initiates is unclear. Radial corallites are budded off in a regular fashion during growth; however, inspection of a 3D model indicates that a new axial corallite wall forms by extension of an existing radial corallite wall, suggesting the radial corallite is 'switched on' to extend rapidly and develop into an axial corallite (Fig. 3a, b).

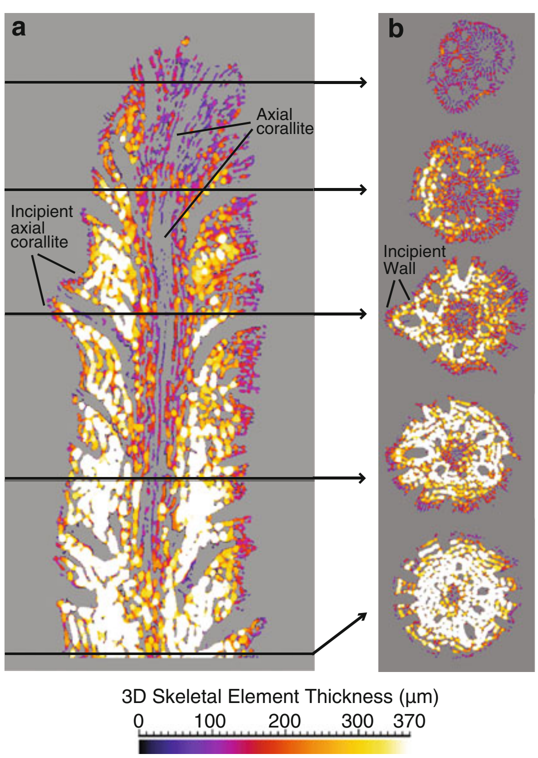


Fig. 3 Coral skeletal element thickness varies along branches of *Acropora pulchra*. Micro-CT reconstruction of specimen 2c with (**a**) axial and (**b**) five transverse sections. The data have been colourised according to the thickness of skeletal elements: *White > Yellow > Orange > Red > Purple > Blue*

As demonstrated here, there are evident advantages to utilising micro-CT scanning technologies for examination of porosity within coral skeleton. Micro-CT imaging allows information on porosity to be obtained with simultaneous spatial imaging of the actual skeleton responsible for variations in measurements, facilitating a better understanding of the processes that drive porosity variations in coral skeleton. Generation of 3D images of internal skeletal structure could also assist in taxonomic identification of coral specimens and has an undoubted application for the non-destructive analysis of museum and fossil specimens.

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